Predicting Pressure Dynamics from Heat Signals via Stacked Vanilla Recurrent Networks

Prioritizing Architectural Simplicity in High-Frequency Modeling

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Part 1

Introduction & Problem Framing

Research Objective & Dynamical Challenges

Objective: Correlate heat release to pressure signals and predict pressure from unseen heat input signals.

Core Dynamical Modeling Challenges:

1. **Data Volume & Sensor Noise:** Excessive raw sequence points causing training overhead.
2. **Cold-Start Transients:** Neural network hidden states initialized at zero cause severe prediction lag at the start of inference windows.
3. **Sequence Memorization:** Deep networks often memorize chronological timelines rather than learning the underlying continuous physical manifold.

Challenge 1: Data Volume & High-Frequency Noise

Temporal Downsampling (`STEP_SIZE = 2`)

Processing every single micro-second of the sensor data leads to redundant calculations. We apply a uniform temporal decimation strategy.

Dual-Purpose Downsampling

- **Primary Purpose (Computational):** Actively reduces the overall dataset size by 50%. This drastically shortens the sequence length for Backpropagation Through Time (BPTT), significantly accelerating training convergence.
- **Secondary Purpose (Physical):** Acts as a low-pass filter. High sampling rates inherently introduce small-scale sensor perturbations; skipping steps strips this redundant noise, preventing the model from fitting non-physical artifacts.

The Initialization Problem ($h_0 = 0$)

By default, recurrent neural networks initialize their internal memory (hidden states, h_t) to absolute zero at the beginning of any new sequence.

The Physical Consequence

If a target prediction window starts at a moment of extreme high pressure, a network starting with $h_0 = 0$ is completely blind to the system's current momentum.

This forces the network to spend the first 100+ time steps "waking up" and trying to find its place on the wave, resulting in massive transient errors at the beginning of every prediction window.

Challenge 3: Sequence Memorization

The Problem: Overfitting to continuous time-series trajectories. The network risks memorizing the chronological timeline rather than learning the underlying continuous physical manifold.

Proposed Interventions:

- **Stochastic Windowing:** Ingesting randomized sequence slices during training. Shuffling the window starting indices completely breaks the chronological sequence, forcing the model to learn local dynamical mappings rather than global trajectory memorization.
- **Architectural Regularization:** Utilizing a high inter-layer dropout rate (40%) across the stacked hidden units continually perturbs the computational graph, heavily penalizing parameter co-adaptation and strict memorization paths.

Part 2

Methodology: Data Pipeline

Data Pre-Processing: Standard Scaling

Target Isolation & Physical Inversion

- High-frequency sequences are ingested from `.txt` files.
- Heat release is multiplied by -1 to establish the correct physical orientation.
- Pressure is isolated as the strict ground-truth target.

Z-Score Normalization

Features and target signals are scaled using standard normalization (`StandardScaler`):

$$z = \frac{x - \mu}{\sigma}$$

This centers the continuous time-series data at zero mean ($\mu = 0$) and scales it to unit variance ($\sigma = 1$), ensuring optimal gradient flow through the network's tanh activation functions without introducing artificial signal boundary bounds.

Feature Engineering: The 2D Tensor

Why Instantaneous Heat is Insufficient:

Feeding only the current heat (H_t) forces the network to guess the trajectory. It knows the current magnitude, but does not know if the wave is currently accelerating upwards or collapsing.

Discrete Difference Operator

We explicitly engineer the discrete derivative to provide the physical trajectory:

$$\text{Instantaneous Magnitude: } H_{\text{scaled}}$$

$$\text{Rate of Change (Acceleration): } \Delta H_t = H_t - H_{t-1}$$

The Final 2D Input Tensor (X_t):

$$X_t = \left[H_{scaled}, \ \ \Delta H \right]$$

Data Preprocessing Tabular Representation

Parameter	Specification	Function
Training Sequence (W_{train})	1500 Timesteps	Batch slice length for sequence training.
Context Buffer (Warmup / Burn-in)	500 Timesteps	Pre-pended history fed into the network to warm up hidden states (h_t) before prediction.
Down Sampling	Step Size $= 2$	Low-pass filter removing non-physical high-frequency noise.
Input Feature Vector (X_t)	2 Dimensions	$[H_{scaled}, \Delta H]$
Normalization	Z-Score Scaling	Prevents activation saturation (tanh) and preserves variance of acoustic peaks.

Part 3

Methodology: Network Design

Empirical Ablation Study: Architecture Selection

Testing on the `FREI` datasets was conducted to determine the optimal recurrent formulation.

Model	Avg RMSE ↓	Avg R^2 ↑	Avg PCC ↑	Key Observation
A: Base RNN (1 layer)	0.0081	0.9752	0.9876	Strong baseline, but lacks depth for non-linear mappings.
B: Stacked RNN (4 layers) (40% Dropout)	**0.0063**	**0.9857**	**0.9934**	**Optimal; More layers resulted in the lowest error.**
C: Base LSTM (1 layer)	0.0100	0.9629	0.9814	Underperformed the Base RNN; gating added unnecessary complexity and over- parameterization.
D1: Stacked LSTM (20% Dropout)	0.0110	0.9556	0.9807	Struggled with overfitting; 20% dropout rate provided insufficient regularization.
D2: Stacked LSTM (40% Dropout)	0.0088	0.9710	0.9877	40% dropout Improved generalization over Model D1 results in the reduction in overfitting.
E: Stacked LSTM (Peak-Loss)	0.0117	0.9503	0.9749	Poorest fit; sacrificed global baseline accuracy just to fit transient spikes.

Primary Architecture: 4-Layer Stacked Vanilla RNN

Because our feature engineering (ΔH) explicitly encodes the short-term physical trajectory, complex long-term memory gates (LSTMs) proved unnecessary. We rely on a deeply stacked Elman Recurrent Network.

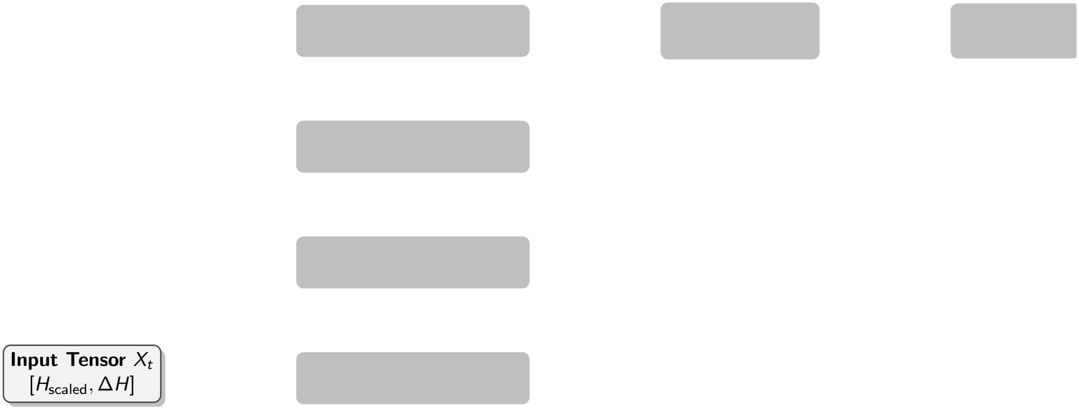
Mathematical Formulation (Per Layer $l \in \{1, 2, 3, 4\}$)

$$h_t^{(l)} = \tanh\left(W_{ih}^{(l)} x_t^{(l)} + b_{ih}^{(l)} + W_{hh}^{(l)} h_{t-1}^{(l)} + b_{hh}^{(l)}\right)$$

Architectural Specifications:

- **Deep Hierarchy:** 4 distinct layers, each with 128 hidden units, allowing the model to combine basic gradients into complex macroscopic wave structures.
- **Latent Projection:** The 128D output of Layer 4 is passed through an Information Bottleneck (`FC1 + ReLU`, 64 units) to strip final noise before projecting to a 1D Pressure scalar (`FC2`).

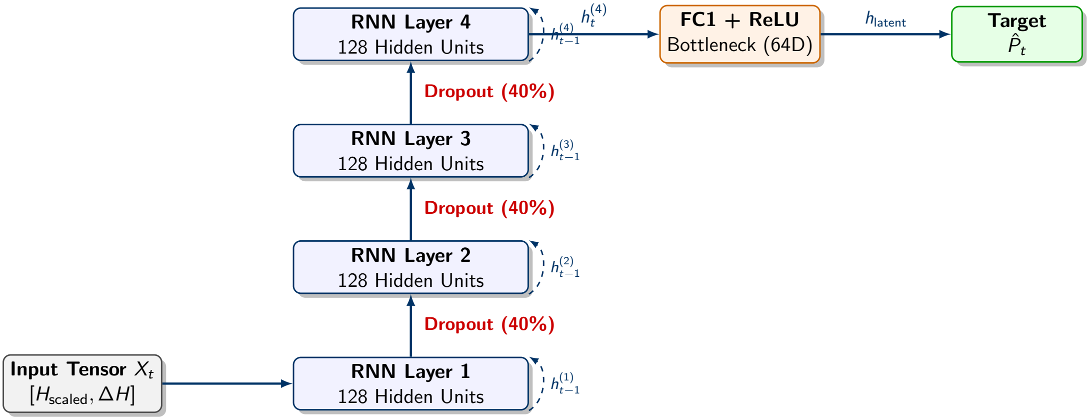
Information Flow: Forward Pass & Backpropagation



Step 1: Input tensor is prepared.

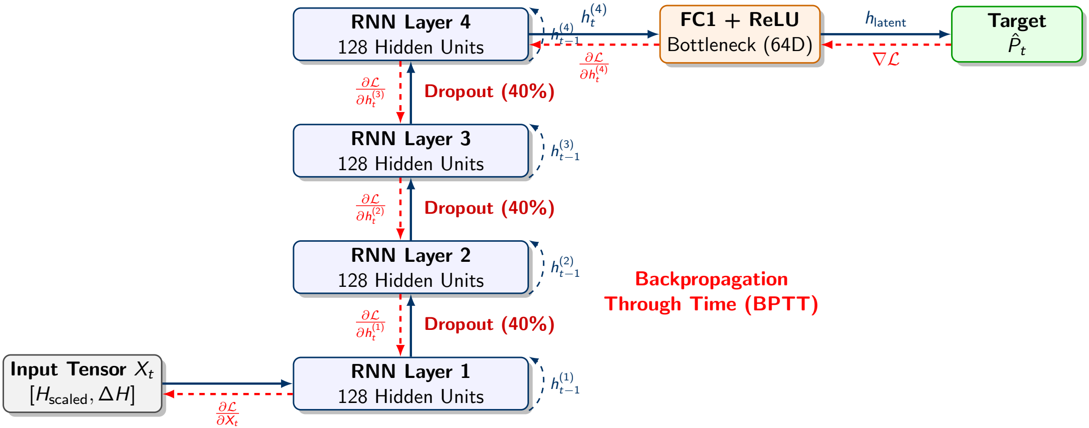
Note: Continuous animation requires Adobe Acrobat Reader to view properly.

Information Flow: Forward Pass & Backpropagation



Step 1: Forward Pass & Recurrent State Updates. Input data and previous hidden states (h_{t-1}) are processed simultaneously to predict pressure dynamics.

Information Flow: Forward Pass & Backpropagation



Step 2: Backpropagation Through Time. Gradients ($\nabla \mathcal{L}$) flow backwards to update weights, calculating partial derivatives at each depth.

Model Architecture Tabular Representation

Parameter	**Specification**	**Function**
Recurrent Core	4-Layer Stacked Vanilla RNN (tanh)	Deep feature extraction for micro-dynamics.
Layer Dimensions	128 Hidden Units	Dimensionality of the hidden memory state (h_t).
FC1 Compression	128D $\rightarrow$ 64D (ReLU)	Latent space information bottleneck layer compressing hidden features down to salient variables.
FC2 Output	64D $\rightarrow$ 1D Scalar	Linear transformation projecting compressed latent features down to a continuous pressure prediction ($\widehat{P_t}$).

Part 4

Methodology: Training & Regularization

Solving Cold-Starts: The Warm-Up Phase

To eliminate the initialization lag caused by $h_0 = 0$, we engineered a dynamic **Warm-Up Buffer** directly into the data loader and training loop.

The 2000-Step Ingestion Strategy

1. **Oversampling:** Instead of extracting a 1500-step window, the dataset class extracts a 2000-step continuous slice.
2. **Burn-in Phase (Steps 0 to 500):** The model processes the first 500 steps. We **do not** calculate loss here. This phase exists purely to populate the hidden state matrices (h_t), allowing the network to physically align its momentum with the real wave.
3. **Target Phase (Steps 501 to 2000):** The loss function (`PeakWeightedLoss`) is strictly applied to the final 1500 steps, completely eliminating cold-start transient penalties.

Regularization: Preventing Memorization

With a 4-layer deep architecture, preventing overfitting to the training timeline is critical.

1. **Heavy Inter-layer Dropout ($p = 0.40$)**
   - Between every stacked RNN layer, 40% of the hidden state vector connections are randomly zeroed out.
   - **Justification:** Testing proved 20% was insufficient. At 40%, we actively shatter neuron co-adaptation, forcing independent feature learning rather than rote memorization.
2. **Stochastic Windowing**
   - Batches are built by randomly slicing the dataset rather than reading it chronologically ($S = \{X_\tau\}_{\tau=k}^{k+W}$).
   - **Justification:** Destroys the model's ability to map predictions to an absolute time index.

Training & Loss Configuration Tabular Representation

Parameter	Specification	Function
Dropout Rate	$p = 0.4$ (40%)	Zeroes hidden units between RNN layers to prevent neuron co-adaptation and overfitting.
Optimizer	Adam (LR = 0.001)	Adaptive moment estimation for momentum-based gradient descent.
LR Scheduler	ReduceLROnPlateau (Factor=0.5, Pat=5)	Halves the learning rate when validation loss plateaus to fine-tune convergence.
Batch Size	8 Sequences	Balances memory efficiency with stochastic gradient noise during sequence training.

Part 5

Results & Validation

Validation in the Time Domain: Full Data

Generative Inference Evaluation

The Model weights are frozen (`torch.no_grad()`) using the optimal learned parameters. The model relies on purely open-loop generation; it maps the full sequence of unseen thermal inputs to pressure dynamics using only its learned internal physics, completely independent of ground-truth feedback.

Metrics Achieved:

- **Avg RMSE:** 0.0063
- **Avg R^2:** 0.9857
- **Avg PCC:** 0.9934

The near-perfect Pearson Correlation Coefficient (PCC) confirms that our feature engineering and warm-up phase successfully eliminated phase lag.

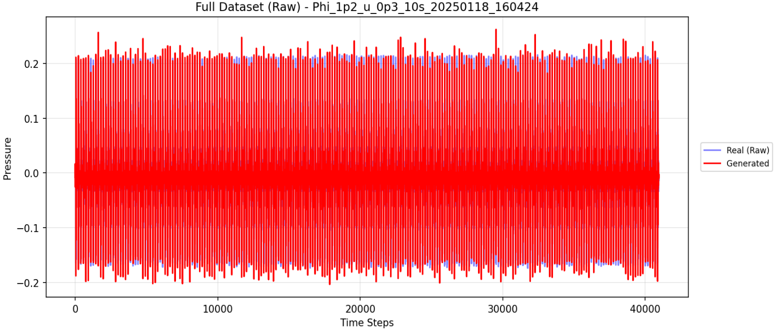


Fig: Continuous generative pressure output vs empirical ground truth.

Validation in the Time Domain: Uniform Window Sampling

Evaluating Generalization Across the Sequence

To rigorously evaluate the model's stability, we extracted six uniformly spaced inference windows (W_1 to W_6) across the entire unseen dataset using a linear space distribution. Each window represents 1500 time steps of continuous open-loop prediction (strictly evaluated after the 500-step warm-up phase). This confirms that generative accuracy is consistent and independent of specific starting regimes.

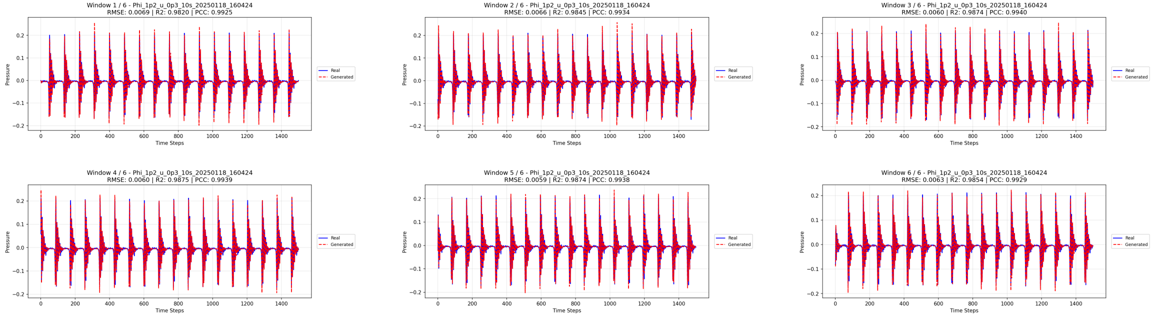


Fig: 1500-step pressure predictions across six distinct temporal locations.

Objective: Statistical metrics alone are insufficient to prove dynamical generalization. We compare the physical attractor to the latent projections to verify if the network learned the true system topology.

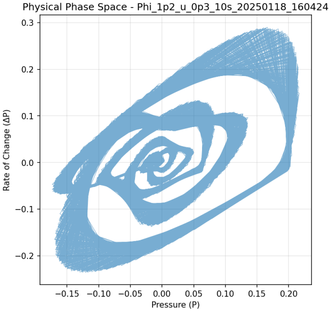


Figure 1: Physical Phase Space (P vs ΔP)

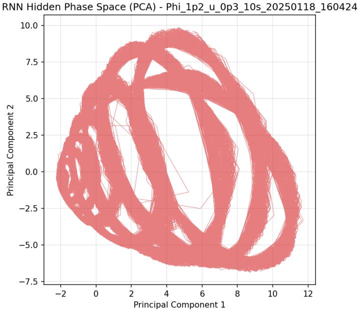


Figure 2: Hidden Phase Space (PCA $h_t^{(4)}$)

Topological Validation: Key Observations

Latent Space Projection (Ref: Figure 2)

The RNN's 128-dimensional internal memory organically projects into a smooth, bounded 2D envelope that directly captures the topology of the thermoacoustic system.

Dynamical Mapping (Figure 1 vs. Figure 2)

This structural similarity proves the model has not merely memorized sequential data points. The hidden state dynamics (**Figure 2**) have organized themselves to perfectly mirror the physical ground truth limit cycle (**Figure 1**).

Continuous Motion

The absence of fractured trajectories in both spaces guarantees that the learned dynamics remain stable and cohesive, maintaining physical realism even while generating over 1500 steps in a strictly open-loop regime.

Latent Temporal Determinism

To further validate open-loop stability, we apply Takens' Embedding Theorem to the hidden state across sequential time delays: $h(t)$, $h(t+1)$, and $h(t+2)$.

Observations from Figure 3:

- **Manifold Unfolding:** The time-delay embedding unfolds into a continuous 3D structure, showing a clear central void characteristic of periodic oscillatory motion.
- **Trajectory Smoothness:** State transitions are gradual and cohesive. This rules out chaotic mode-hopping or exploding gradients within the vanilla RNN.
- **Intrinsic Dimensionality:** Despite living in a 128D space, the dynamics strictly collapse onto this tight manifold, meaning the model is operating purely on the learned physical rules rather than noise.

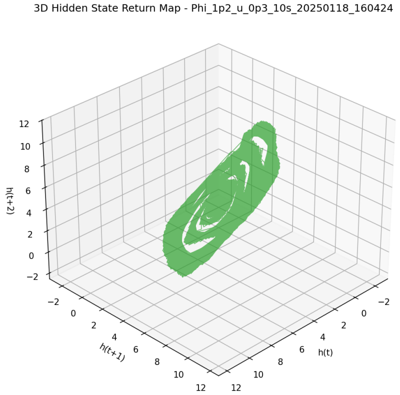


Figure 3: 3D Hidden State Return Map

Part 6

Conclusion & References

Conclusion

Summary of Findings:

1. **Architectural Efficiency:** A deeply stacked (4-layer) Vanilla RNN outperforms gated LSTM networks for this specific class of pressure prediction, mapping sequential dynamics effectively.
2. **Dynamic Feature Dominance:** Providing ΔH as an explicit feature eliminated the need for complex long-term recurrence mechanisms.
3. **Initialization Stability:** Implementing a 500-step Warm-up Buffer successfully stabilized hidden state matrices, completely resolving initial phase errors.

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Thank You for Your Attention!

Any Questions?